FIRST SEMESTER M.Sc. DEGREE EXAMINATION, NOVEMBER 2018

(CUCSS—PG)

Mathematics

MT 1C 03—REAL ANALYSIS—I

(2016 Syllabus Year)

Time : Three Hours Maximum : 36 Weightage

Part A (Short Answer Questions)

Answer **all** *questions.*
Each question carries weightage 1.

1. Prove that the neighbourhood of a point in a metric space is an open set.
2. If X is a metric space and $E \subset X$, prove that $\bar{E} \subset F$ for every closed set $F \subset X$ such that $F \supset E$.
3. Show that the set of limit points of a bounded set is bounded.
4. Let X be a compact metric space. Is the continuous image of X in $\mathbb{R}$ closed ? Justify.
5. Examine the continuity of the function f, at the origin, defined by $f(x) = \begin{cases} x \sin \frac{1}{x} \text{ if } x \neq 0 \\ 0 \text{ if } x = 0 \end{cases}$.
6. Prove that every uniformly continuous function in an interval is continuous in that interval. Is the converse true ? Justify.
7. If f is differentiable in (a, b) and $f'(x) \geq 0$ for all $x \in (a, b)$, prove that f is monotonically increasing.
8. Prove that $\lim_{x \to 0} x \cos\left(\frac{1}{x}\right) = 0$.
9. Check whether the Mean Value Theorem is applicable to $f(x) = 2 + (x-1)^{\frac{2}{3}}$ in $[0, 2]$.
10. Define Riemann Stieltjes integral of f with respect to α over $[a, b]$. Illustrate it with an example.

11. For any partitions P_1 and P_2 of $[a, b]$, prove that $L(P_1, f, \alpha) \leq U(P_2, f, \alpha)$.

12. If γ is the curve in $\mathbf{R}^3$ defined by $x(t) = 2t - 1, y(t) = t + 1, z(t) = t - 2$, show that γ is rectifiable.

13. If the family $\{f_n\}$ of continuous functions defined on a compact metric space K is uniformly continuous, prove that $\{f_n\}$ is equicontinuous on K.

14. Prove that every member of an equicontinuous family of functions is uniformly continuous.

(14 × 1 = 14 weightage)

Part B

Answer any **seven** *questions from the following ten questions.*
Each question carries weightage 2.

15. Let $Y \subset X$. Prove that $E \subset Y$ is open relative to Y if and only if $E = Y \cap G$ for some open set G of X.

16. Let A be a compact metric space and $f : A \to \mathbf{R}^k$ be continuous. Prove that f attains its bounds.

17. Let $k \in \mathbf{N}$. If $\{I_n\}$ is a sequence of k-cells such that $I_n \supset I_{n+1}, n = 1, 2, \ldots$, prove that $\bigcap_{n=1}^{\infty} I_n$ is not empty.

18. If E be a non-compact set in $\mathbf{R}$ then show that there exists a continuous function on E which is unbounded.

19. Let f be defined by $f(x) = x^2$. Is f uniformly continuous on R ? Justify your answer.

20. If $f(x) = |x|^3$, compute $f'(x), f''(x)$, for all real x and show that $f^{(3)}(0)$ does not exist.

21. If $f \in \mathcal{R}(\alpha)$ on $[a, b]$, $m \leq f \leq M$, ϕ is continuous on $[m, M]$ and $h(x) = \phi(f(x))$ on $[a, b]$, prove that $h \in \mathcal{R}(\alpha)$ on $[a, b]$.

22. If $f \in \mathcal{R}(\alpha_1)$ and $f \in \mathcal{R}(\alpha_2)$, prove that $f \in \mathcal{R}(\alpha_1 + \alpha_2)$ and find $\int_a^b f d(\alpha_1 + \alpha_2)$.

23. State and prove the Cauchy's criterion for uniform convergence of sequence of functions defined on $E \subseteq R$.

24. If $f_n \to f$ uniformly on $[a, b]$, $f_n \in \mathcal{R}(\alpha)$, on $[a, b]$, where α is monotonically increasing on $[a, b]$, prove that $f \in \mathcal{R}(\alpha)$ on $[a, b]$ and $\int_a^b f d\alpha = \lim_{n \to \infty} \int_a^b f_n d\alpha$.

(7 × 2 = 14 weightage)

Part C

Answer any **two** *questions from the following four questions.*
Each question carries weightage **4**.

25. (a) Let $f : X \to Y$ be continuous where X and Y are metric spaces. If E is a connected subset of X, prove that $f(E)$ is connected.

(b) Let X and Y be metric spaces. Prove that $f : X \to Y$ is continuous if and only if $f^{-1}(C)$ is closed in X for every closed set C in Y.

26. Suppose α increases monotonically, $\alpha' \in \mathcal{R}$ on $[a, b]$ and f is a bounded real function on $[a, b]$. Prove that $\int_a^b f d\alpha = \int_a^b f(x) \alpha'(x) dx$.

27. Find a real continuous function on the real line which is nowhere differentiable.

28. State and prove the Stone-Weierstrass theorem.

(2 × 4 = 8 weightage)